

FLATTRIKE · CALCULATION NOTE

FlatTrike sizing and first-principles checks

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DOCUMENT CONTROL

Document ID	FTK-CAL-001	Version	0.1
Status	Draft	Date	2026-09-25
Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/flattrike		
Source	docs/04-calcs/01-sizing.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.1	2026-09-25	Amish Chadha	First TRL 3 sizing note (geometry, nesting, mass, power, stability, structure, braking, assembly, cost) against every requirement

FlatTrike sizing and first-principles checks

On paper the flat-plate frame is strong enough in bending, but three structural features of the TRL 2 concept would have failed and are changed in the TRL 3 model: the open twin-plate spine (torsion), the flat fork bridges (bending) and the narrow stay strips (buckling). With those changes, eight of the eighteen requirements are met, three are at risk and one cannot be verified at TRL 3. Six are **not met**: R1 with a 90 kg rider (307.5 kg gross), R8 (empty mass 67.5 kg against 55 kg), R10 (rider-only tipping at 0.24 g, and 0.11 g at full box-steering lock), R13 without assist (238 W needed at the pedals), R14 for the spine side plates (about 78 min to replace), and R18 (about \$414 per trike at 100 units against \$300).

Every number in this note is printed by `docs/04-calcs/sizing.py` (run from the repo root: `python docs/04-calcs/sizing.py`), which also writes `docs/04-calcs/results.csv`. The script reads plate areas, part centroids and key dimensions from the parametric model `cad/src/model.py`, and prices from `bom/bom.csv`, so the model, the drawing FTK-DWG-001 and this note agree. All values are first-principles estimates; nothing here is measured.

■ 1. Assumptions

Table 1. Inputs. All are assumptions for a paper design unless marked as decided.

Input	Value	Basis
Layout and steering	Tadpole front loader, box steering on one kingpin, 35° lock	Decided, FTK-DDR-001 items 1 and 2; lock angle assumed from bed-to-frame clearance
Rated load	150 kg cargo, 80 kg rider (90 kg for the upper case)	FTK-REQ-001 R1
Plate	3 mm S235JR or A36, 7,850 kg/m ³ (23.6 kg/m ²), yield 235 MPa, E = 210 GPa	Decided, item 4
Plywood	700 kg/m ³ ; 12 mm floor, 9 mm walls and lid	Exterior grade; TRL 3 choice within the TRL 2 range
Rider centre of mass	20 mm ahead of the saddle, 160 mm above its top: 0.42 m ahead of the rear axle, 1.09 m high	Upright seated rider; saddle on a 73° seat line
Cargo centre of mass	Centre of the box interior: 1.46 m ahead, 0.59 m high	Uniform load
Bought part masses	Front wheels 5.8 kg (pair), rear wheel 3.8, drivetrain 3.0, headset and head tube 1.3, seat 1.4, bars 1.2, brakes 0.7, box hardware 1.0, paint 0.6, accessories 1.5 kg	Typical steel-rim utility parts
Fasteners	112 M8 class 8.8 bolt sets at 28 g; 72 cut spacer washers	Joint count in sizing.py ; all-metal locknuts decided (item 5)
Riding	Crr 0.015, CdA 0.9 m ² , air 1.2 kg/m ³ , drivetrain 89 %, rider 110 W sustained	As TRL 2
Gearing	32T chainring, 24T sprocket, hub ratios 0.75, 1.00 and 1.33	Typical 3-speed hub
Dynamic factor	2.5 on static loads for potholes and curbs	As TRL 2; EN 17860 test loads not read (see section 10)
Bolted joints	Preload 15.6 kN at 25 N·m (K = 0.2); slip factor 0.20 on painted faces	Typical
Fatigue	Plate with holes treated as a 90 MPa detail, constant-amplitude limit 66 MPa	Eurocode 3 practice for members with holes; to confirm
Drums	0.35 kg iron drum and 0.30 kg aluminium shell per hub; heat loss 0.5 W/K at 10 km/h	Assumed; no hub data read
Assist (route C only)	250 W mid-drive, 80 % motor and controller, SwapCell 457 Wh per cycle at the terminals	SWC-CAL-001
Production (R18)	One full sheet at \$1.0/kg, cutting at \$0.30/m, bought parts at 65 % of prototype retail, plywood at 60 %, \$15 assembly labour	Assumed volume rates; no quotes

■ 2. Geometry, plates and nesting

The parametric model is 2.11 m long, 0.986 m wide and 0.93 m high to the saddle, on a 1.45 m wheelbase and 0.84 m track. It has 27 cut plates with a net area of 1.22 m² (28.6 kg), plus 72 spacer washers cut from the offcuts, so every steel part is still flat 3 mm plate (R3). The head tube and bottom bracket shell are bought steel sections (decided, item 6).

A first-fit shelf nest of the plates' bounding rectangles, with 8 mm gaps and 10 mm edge margins, uses 2,486 mm of a 1,250 x 2,500 mm sheet (R4). This is conservative, because true-shape nesting interlocks the spine and stay plates, but the margin is only 14 mm, so any plate growth must be checked against the DXF nest. The plates use 38.9 % of the sheet by net area.

The box holds 184 L with the lid (counter) at 0.77 m (R17). Plates and box panels stack about 172 mm deep on a 0.80 x 0.80 m footprint; with the three wheels stacked beside them the crate is about 1.35 x 0.80 x 0.36 m, or 0.39 m³ (R6). The longest cut plate is the 800 mm bed rail and the longest piece is the 795 mm box floor.

■ 3. Mass and centre of mass

The empty trike is **67.5 kg** against the 55 kg target (R8 **not met**). Plates are 42 % of it (28.6 kg) and the box 22 % (14.7 kg); bolts and washers add 3.9 kg. The TRL 2 figure of about 68 kg was close by chance: the lightening windows saved about 6 kg, but the TRL 2 box mass was low by about 3 kg and the new cover and crown plates add weight.

Table 2. Mass and axle loads.

Quantity	Value
Empty mass	67.5 kg
Gross, 80 kg rider and 150 kg cargo	297.5 kg (R1 met, 2.5 kg margin)
Gross, 90 kg rider and 150 kg cargo	307.5 kg (R1 not met)
Cargo allowed with a 90 kg rider inside 300 kg	142.5 kg
Empty trike centre of mass	0.98 m ahead of the rear axle, 0.47 m high
Axle loads, loaded	front 2.16 kN, rear 0.76 kN

The loaded trike carries about three quarters of its weight on the front wheels, so the single driven rear wheel is lightly loaded (section 4).

■ 4. Riding power, gearing and assist

At 297.5 kg gross, rolling resistance is 43.8 N and 110 W at the pedals gives 97.9 W at the wheel, for a flat cruise of **7.6 km/h** (R12 met): 92.8 W goes to rolling and 5.1 W to drag (media Figure: [media/flow.png](#)).

A 5 % grade needs 189.7 N, which is 211.5 W at the wheel and **238 W at the pedals** at 4 km/h, more than twice sustained rider output (R13 **not met** unassisted). On 110 W the trike climbs at only 1.9 km/h. The rear tyre needs a grip coefficient of 0.25 to drive up the grade, which is adequate on dry asphalt but at risk on wet or loose surfaces.

With a 32/24 drive and the hub's 0.75 low ratio, the development is 1.60 m per crank turn in low gear (2.13 and 2.84 m in middle and high). At 4 km/h in low gear the cadence is 42 rpm, the mean crank torque 54 N·m and the mean pedal force 319 N; in high gear at 70 rpm the trike does 11.9 km/h. The hub maker's input torque limit must be checked for this low gearing.

With assist route C (a 250 W mid-drive on a SwapCell pack, FTK-DDR-001 item 3), the loaded trike climbs 5 % at 6.0 km/h, which meets R13. It uses about 8.0 Wh/km on the flat at 12 km/h, about 57 km per SwapCell pack, and about 52 Wh/km on a 5 % climb.

■ 5. Steering, turning and stability

At 35° of box-steering lock the outer tyre sweeps a **5.67 m** turning circle and the box corner 5.53 m (R9 met, with the 0.986 m width 14 mm inside the 1.0 m limit). At full lock the outer front wheel moves 334 mm inward, because the kingpin is 450 mm behind the front axle.

The tipping threshold is the distance from the centre of mass to the axis through the rear contact and the outer front contact, divided by the height of the centre of mass. The script moves every bed-mounted mass (bed plates, box, cargo, front wheels, handlebar) with the steering angle.

Table 3. Tipping thresholds (R10 target 0.30 g in every case).

Case	Straight	Box steering, full lock	Option B (Ackermann), any lock
Loaded, 80 kg rider	0.43 g (16.6 km/h on a 5 m radius)	0.31 g (tips at 9.8 km/h)	0.43 g
Rider only, 80 kg	0.24 g (12.2 km/h on a 5 m radius)	0.11 g (tips at 5.8 km/h)	0.24 g
Rider only, 90 kg	0.23 g		

R10 is **not met** with no cargo. Moving the seat forward to a 73° seat angle raised the rider-only threshold from the TRL 2 value of 0.18 g to 0.24 g, but closing the gap needs one of: the rider 350 mm further forward (not practical with a rear-drive layout), 30 kg of ballast low in the box, or a wider track (0.25 g at 0.90 m and 0.28 g at 1.0 m, both breaking R9). The more serious finding is box steering at full lock: an empty trike tips at 5.8 km/h in its tightest turn. Ackermann steering (option B) keeps the track constant and removes that case. This is proposed for the first prototype, awaiting Amish (FTK-DDR-001 open item 13).

■ 6. Structure (R2)

Bending stresses in the plates are low. The TRL 3 checks found that stability, torsion and joint behavior govern, which is what ITDP found in traditional rickshaws. The rear frame and rider (112 kg) hang on a single rear wheel, so any roll moment passes through the kingpin into the spine as torsion.

Table 4. Structural checks (dynamic factor 2.5 unless stated).

Check	TRL 2 concept	TRL 3 model	Result
Spine torsion, 0.5 g roll (501 N·m)	556 MPa shear in two open 3 x 150 mm plates	9.0 MPa in a closed 63 x 150 mm box with top and bottom covers	Fixed by the covers; GJ rises from 219 to about 196,000 N·m ²
Cover tabs, shear flow 27 N/mm		84 MPa bearing, 10 tabs per edge over 280 mm	Acceptable; tab count to confirm in FEA
Fork crown, 1.35 kN outer fork plate load	619 MPa in a flat 160 x 3 mm bridge	23 MPa in two vertical 80 mm crown plates	Fixed by the crown plates
Upper stay strip, 2.53 kN compression per plate	SF 1.13 with a 55 mm strip braced at 300 mm	SF 2.06 with a 100 mm strip braced at 300 mm (SF 0.50 if the stay bridge were omitted)	Marginal; buckling governs the stays
Front axle beam, 5.40 kN on a 730 mm span		45 MPa (twin 3 x 110 mm plates with windows)	Met; 24.5 MPa under the R2 proof load of 2.94 kN
Spine bending at the stay front node		0.7 MPa; the bed carries almost no vertical load into the kingpin (-2 N loaded, 39 N empty)	Negligible
Headset under roll		3.34 kN radial on each bearing (150 mm spacing)	At risk: a bicycle headset is not designed for this moment
Stay-to-spine nodes, 2.53 kN per plate		Slip load 3.12 kN per bolt per face; 2 bolts per plate per node; bearing use 7 % if slipped	Met on paper if preload is kept
Fatigue at M8 holes		Net stress range 3.7 MPa per road cycle and 11 MPa per pothole, against a 66 MPa limit; about 3 million cycles in 5 years	Met on paper; fretting and slip not assessed

R2 is **at risk**: the nominal numbers pass, but the stay strips have only about twice the buckling load, the headset carries a roll moment it was not designed for, and joint slip, fretting and the EN 17860 load cases are unassessed. A plate-and-joint FEA is the next step on paper.

■ 7. Braking and parking (R11)

Stopping from 15 km/h in 6 m needs 1.45 m/s² (2.22 m/s² if 0.5 s of reaction is counted inside the 6 m), from 2.58 kJ of kinetic energy. Shared equally, each drum must give about **36 N·m**, and the front pair alone needs a tyre grip coefficient of only 0.20. Parking on a 10 % grade needs 290 N, or **37 N·m** per front drum with the latch on the front pair. The drum torque rating of the chosen hubs has not been read, so these are requirements on the hub, not checks.

Holding 10 km/h down a 5 % grade puts 272 W into the drums. With the assumed drum heat capacity (431 J/K each) and cooling (0.5 W/K), the steady rise would be 182 K, and the drums reach a 100 K rise after 11.5 min, or about **1.9 km** of continuous descent. R11 is **at risk** for long descents and cannot be verified at TRL 3 for stopping and parking torque.

■ 8. Assembly, repair and service life (R7, R14, R16)

At 1.5 min per bolt set plus the wheels, headset, drivetrain, brakes, seat, bars, box and a final torque check, assembly is about 5.6 person-hours, or **3.6 h** for two people working 70 % in parallel (R7 **at risk**: under 4 h but with little margin and no timed build).

Replacing a fork plate disturbs 6 bolts and takes about 24 min, but a spine side plate disturbs 42 bolts (covers, ribs, stay nodes, collars and bottom bracket) and takes about **78 min** (R14 **not met** for the spine side plates).

Unprotected 3 mm steel at the upper first-year corrosion rate of ISO 9223 category C3 (50 µm) or C4 (80 µm), with a time exponent of 0.6, loses about 0.26 to 0.42 mm of thickness over 5 years across both faces. The paint system and sealed joint faces must hold that off; R16 **cannot be verified at TRL 3**.

■ 9. Cost (R15, R18)

The pedal-only prototype costs **\$647** in `bom/bom.csv` (15 lines, every line priced), \$153 inside the \$800 budget (R15 met). The optional route C assist kit is \$300 without the SwapCell pack, which is priced once in the SwapCell repo and excluded here (decided, cross-cutting).

At 100 units the estimate is **\$414** per trike: \$74 of steel (one full sheet), \$15 of cutting (about 51 m at \$0.30/m), \$274 of bought parts at 65 % of retail, the plywood box at 60 % and \$15 of assembly labour. R18 (\$300 at 100 units) is **not met**; bought bicycle parts are two thirds of the cost, so the target depends on the partner's wholesale prices more than on the frame.

■ 10. Results against every requirement

Table 5. Requirement status at TRL 3. Not met items first.

ID	Requirement (target)	Value	Status
R1	Gross 300 kg or less with 150 kg cargo and a rider up to 90 kg	297.5 kg (80 kg rider); 307.5 kg (90 kg rider)	Not met at 90 kg
R8	Empty mass 55 kg or less	67.5 kg	Not met
R10	Tipping threshold 0.30 g loaded and rider only	0.43 g loaded; 0.24 g rider only; 0.11 g rider only at full box-steering lock	Not met (rider only)
R13	5 % grade at 4 km/h without dismounting	238 W at the pedals needed unassisted; 6.0 km/h with route C assist	Not met unassisted
R14	Any plate or part replaced in 30 min or less	24 min for a fork plate; 78 min for a spine side plate	Not met (spine side plate)
R18	Production cost \$300 or less at 100 units	about \$414	Not met
R2	Frame strength and fatigue	Stresses 45 MPa or less; stay buckling SF 2.06; headset 3.34 kN per bearing	At risk
R7	Two people, 4 h or less, hand tools	about 3.6 h	At risk
R11	Stop from 15 km/h in 6 m; park on 10 %	36 to 37 N·m per drum needed; fade after about 1.9 km of 5 % descent	At risk
R16	5 years of commercial service	Up to 0.42 mm corrosion loss if unprotected	Not verifiable at TRL 3
R3	Flat 3 mm plate only, bolted	27 plates and 72 washers, all flat	Met
R4	One 1,250 x 2,500 mm sheet	2,486 mm used by bounding rectangles	Met (14 mm margin)
R5	Standard bicycle parts	20 in ISO 406, 1 1/8 in headset, 68 mm BB, 27.2 mm post	Met by design; availability survey pending
R6	0.4 m ³ or less; no piece over 1.0 m	0.39 m ³ ; longest piece 0.80 m	Met
R9	Width 1.0 m, length 2.2 m, turning circle 6 m or less	0.986 m, 2.11 m, 5.67 m	Met
R12	7 km/h on the flat at 110 W	7.6 km/h	Met
R15	Pedal-only prototype \$800 or less	\$647	Met
R17	150 L box; counter at 0.75 to 0.95 m	184 L; 0.77 m	Met

Counts: 8 met, 6 not met, 3 at risk, 1 not verifiable at TRL 3.

■ 11. Limits of this note

- EN 17860 test loads and cycle counts were not read. A public summary confirms the series covers multi-track cargo cycles up to 300 kg and that commercial use doubles the pedaling-force test cycles ([ACT Lab](#)); the numeric load cases need the standard itself.
- Stress checks are hand calculations on simplified members. A plate-and-joint FEA of the spine box, stays and bed is needed before any build decision.

- Hub torque ratings, headset bearing ratings, drum heat data and laser-cutting prices are assumptions until supplier data or quotes are obtained.

Safety: These are paper estimates. Nothing in this note shows the frame is safe to ride. The empty trike can tip at walking pace at full box-steering lock, the headset carries an unusual roll moment, and drum brakes may fade on long loaded descents.